Jiacheng Wang

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Electrical Engineering Co-op Student | Seeking Fall 2026 Hardware / Embedded / Electrical Co-op

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science, Electrical Engineering (Co-op) — *Sep. 2025 – 2030 (Expected)*

- Relevant coursework: Digital Circuits & Systems, Linear Circuits, Electricity & Magnetism, Discrete Mathematics & Logic, Multivariable Calculus, Programming Principles

Technical Skills

Programming: C++, Python, Java, Verilog HDL
Hardware & EDA: Altium Designer, KiCad, Quartus Prime, PCB layout, schematic capture, soldering, PCB assembly, multimeter
Embedded / Robotics Tools: ROS 2, Docker, Foxglove Studio, Git/GitHub
Design & Simulation: COMSOL Multiphysics, Figma

Engineering & Work Experience

Electrical Team Member — Waterloo Aerial Robotics Group (WARG) — May 2026 – Present
Student Design Team — *Waterloo, ON*

- Designing a 12V-to-3.3V buck-converter power-delivery board using the TPS564208 in Altium Designer, including schematic capture, footprint selection, and PCB layout

Electrical Team Member — Waterloo Rocketry — May 2026 – Present
Student Design Team — *Waterloo, ON*

- Soldered and assembled avionics PCBs from KiCad schematics and BOMs, supporting flight-electronics hardware development
- Verified boards and wiring through continuity checks, battery-voltage measurements, component-spec confirmation, and PCB revision feedback

UX/UI Designer — WE Accelerate Program (Magnify Access Inc.) — Jan. – Apr. 2026
Web Design & Marketing Program — *Waterloo, ON*

- Translated client requirements into Figma prototypes for a multi-role event check-in platform with admin, staff-assisted, and self-service user flows
- Designed interface specifications and workflows for a client-facing accessibility-focused product; received an **Outstanding** co-op work-term evaluation

Projects

Personal LDO Regulator Board (Spring 2026) | *Altium Designer, DFM*

- Designed a 5V-to-3.3V NJM2884 linear regulator board through a full Altium schematic-to-layout flow, applying DFM practices including copper pour, thermal relief, and via sizing
- Completed peer design review and finalized the board design for fabrication readiness

Human-Aware Robotic Arm — Toyota Innovation Challenge 2026 | *Dobot, Orbbec, MediaPipe; Best Prototype Award*

- Built a robotic arm prototype with automatic and manual modes, using depth-camera input and MediaPipe hand tracking to support safer object pickup and handoff
- Implemented proximity-aware behavior that slowed or stopped motion based on operator distance; designed the control UI and task-management workflow

Autonomous Robot Navigation (Spring 2026) | *C++, ROS 2, Docker, Foxglove*

- Built a ROS 2 node that converts raw 2D LiDAR scans into a navigable occupancy-grid costmap with dynamic sizing and obstacle cost falloff
- Visualized navigation data live in Foxglove and packaged the development environment in Docker for reproducible testing

Leadership & Awards

VEX Robotics Competition — 2021 – 2022
Team Captain / Builder / Driver — *Provincial First · National Second*

- Founded and captained a rookie robotics team; contributed to mechanical build, primary driving, and autonomous-routine programming

International Space Settlement Design Competition (ISSDC) — 2021 – 2022
Team Leader — *National Gold · Asia Gold · Int. Silver*

- Led cross-functional subteams and owned structural design work for a competition spacecraft concept